Predicting Cognitive Stress from Keystroke & Mouse Dynamics

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0.704	+0.10	13	464
Random-CV AUC (Random Forest)	improvement over M1 baseline	behavioral features (keystroke + mouse)	labeled 10-min windows

▶ Problem & Motivation

Cognitive stress and fatigue degrade decision-making, accuracy, and well-being — especially in knowledge work where they often go unnoticed. Self-report surveys are intrusive and intermittent.

We ask: **can passive behavioral signals — how a person types and moves their mouse — provide a continuous, non-intrusive estimate of their cognitive state?**

If so, the same signals already captured by every operating system could power lightweight wellness monitoring with no extra hardware.

▶ Datasets

Stress Detection by Keystroke & Mouse (Kaggle)

- 2 users, 27,233 keystroke events, 5.2M mouse events
- 98 self-reported stress / fatigue labels
- Used for supervised training and evaluation

IKDD Keystroke Dynamics (MDPI / GitHub)

- 374 user sessions of raw inter-key intervals
- Demographic metadata across age and gender
- Used for feature distribution validation

▶ Experiments & Model Selection

Features (13 total) extracted on 10-minute sliding windows (50% overlap):

- **Keystroke (9):** dwell mean/std/median, IKI mean/std/CV/median, typing speed, error rate
- **Mouse (4):** movement-speed mean/std, click rate, move rate

Models compared with grid-search hyperparameter tuning:

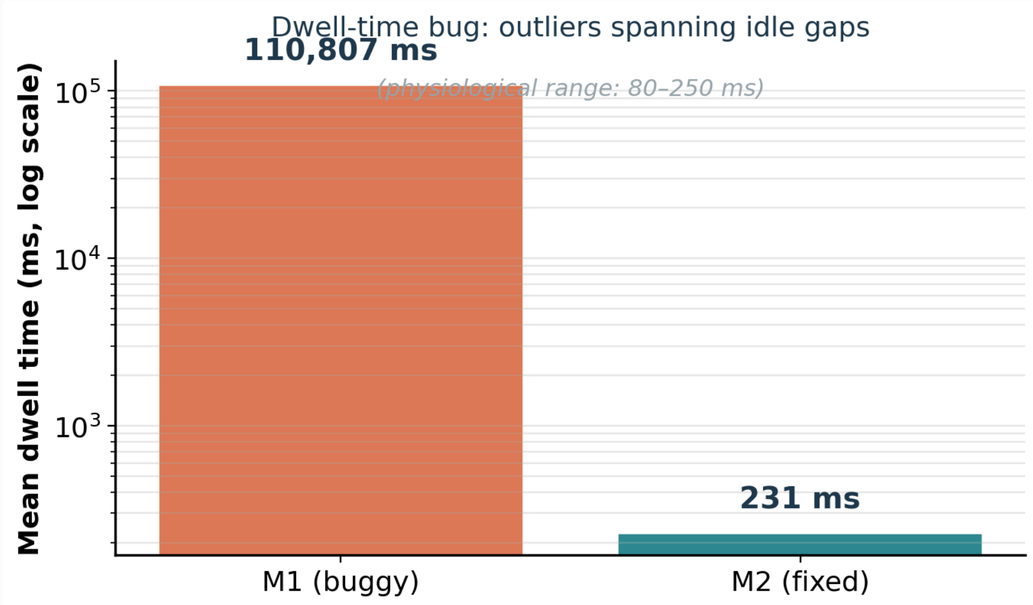
- Logistic Regression (M1 baseline) · Random Forest · Gradient Boosting · SVM-RBF

Two evaluation regimes:

- *Random 5-fold CV* — within-user discrimination
- *Leave-one-user-out CV* — true cross-user generalization

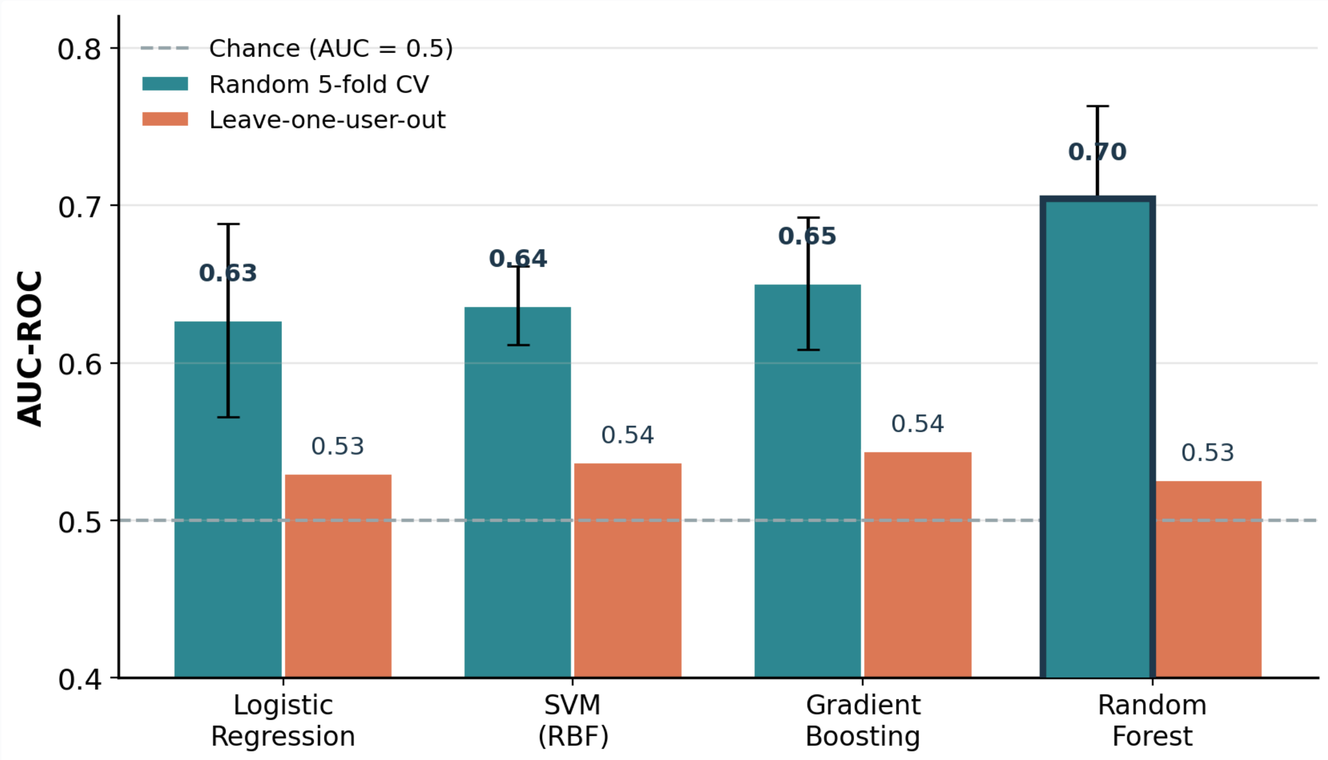
▶ What changed since Milestone 1

Fixed dwell-time outlier bug. M1 reported mean dwell ≈ 110,000 ms; release timestamps spanning idle gaps inflated the differences. After applying the same outlier filter used for inter-key intervals, dwell time falls into its expected ~100–250 ms range.



Also added: mouse features, label-lookback cap (30 min), user-held-out evaluation, ensemble models with hyperparameter tuning.

▶ Evaluation Results



AUC-ROC of each model under random 5-fold CV (teal) and leave-one-user-out CV (coral). Error bars: standard deviation across folds.

Model	Random CV	LOUO
LogReg *(M1)*	0.627	0.530
SVM-RBF	0.636	0.537
Gradient Boosting	0.651	0.544
Random Forest ⭐	**0.704**	**0.526**

Headline finding: Random Forest with mouse features beats the M1 baseline by **+0.10 AUC** on random CV. But all models collapse to **~0.53 AUC** under leave-one-user-out — meaning the apparent signal is largely **per-user typing fingerprint**, not a universal stress signature.

▶ Honest Limitations & Future Work

Limitations

- Only **2 users** in the labeled supervised set — LOUO is effectively 2-fold and statistically underpowered.
- Stress labels are self-reported and intermittent (every ~30–60 min); ground-truth stress could change between reports.
- The labeling join treats stress as piecewise-constant between reports, which is a simplification of the real underlying state.

Future work

- **Per-user normalization** — z-score features against each user's own baseline so the model learns *deviations* from typical typing rather than absolute values.
- **Many more users** — replicate the labeling protocol with ≥ 20 users so cross-user evaluation has statistical power.
- **Live deployment** — the included Streamlit app already supports browser-side feature capture for real-time inference.